Retail HQ + POS Backup Plan

Group: Shade, Sameth, Aisea

Executive Summary

This plan focuses on protecting three main systems: the POS SQL database, product media/files, and the company website. The biggest priority is the POS database because stores depend on it to make sales. If it goes down, the business stops making money.

To meet the required recovery goals, we use a **tiered backup strategy**:

- **POS database** → fastest recovery (replication + frequent backups)
- **Website** → moderate speed recovery
- **File share (media/docs)** → slower, lower-cost backups

We also balance performance with cost by keeping critical data on faster systems and large files on cheaper storage.

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1) Systems & Targets

Data Type	Size (GB)	Daily Change (GB)	RTO (hrs)	RPO (hrs)	Notes
POS SQL Database	300 GB	15 GB	0.5 hrs	0.25 hrs	Most critical system
Website (IIS)	50 GB	2 GB	2 hrs	1 hr	Customer-facing
File Share (media/docs)	2000 GB	20 GB	Next Business Day	24 hrs	Large, not urgent

Assumptions:

- Rounded numbers for simplicity
- SQL is high-transaction / always changing
- File share is large but changes slowly

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2) Backup Plan & Schedule

Data Type	Backup Type	Frequency	Retention	Copies	Verification	Test Restore
POS SQL	Full + log backups + replication	Full weekly, logs every 15 min	1 month	Local + cloud	Daily checks	Monthly
File Share	Full + incremental	Weekly full, daily incremental	1 month	Cloud only	Weekly	Quarterly
Website	Full + incremental	Daily full	2 weeks	Local + cloud	Weekly	Monthly

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3) Storage Estimate (4-week method)

POS SQL

Stored = 4×300 + 24×15
= 1200 + 360 = **1560 GB per copy**
2 copies → **3120 GB**

File Share

Stored = 4×2000 + 24×20
= 8000 + 480 = **8480 GB**

Website

Stored = 4×50 + 24×2
= 200 + 48 = **248 GB per copy**
2 copies → **496 GB**

Total Storage

= 3120 + 8480 + 496 = **12,096 GB (12 TB)**

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4) Cost Estimate

Cloud storage (example: $0.020/GB-month)
12,096 × 0.020 = **$242/month**

Restore egress (example: $0.09/GB)
POS restore (300 GB):
300 × 0.09 = **$27**

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5) Bandwidth & Time Estimates

Formula: Hours = (GB × 2.8) ÷ Mbps

Backup (uplink 50 Mbps)

- SQL daily change (15 GB):
  (15 × 2.8) ÷ 50 = **0.84 hrs (50 min)**
- File share (20 GB):
  1.1 hrs

Restore (downlink 200 Mbps)

- SQL full restore (300 GB):
  (300 × 2.8) ÷ 200 = **4.2 hrs**

This is too slow so it's solved with **replication (failover instead of restore)**

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6) Final Design Decisions

POS SQL (Critical Tier)

- Use **replication or mirroring**
- Backup logs every 15 minutes
- Failover instead of full restore
- Meets:
  - RTO: 30 minutes
  - RPO: 15 minutes

Website (Medium Tier)

- Daily backups + cloud copy
- Quick restore possible

- Meets:
    - RTO: 2 hours
    - RPO: 1 hour

File Share (Low Tier)

- Weekly full + daily incremental
- Stored in cheaper cloud storage
- Meets:
    - RTO: 24 hours
    - RPO: 24 hours

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7) Disaster Playbook

1. Ransomware (today)

- Use last clean backup (cloud)
- Restore SQL to last log backup
- RPO: 15 min
- RTO: 30 min

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2. Deletion after 72 hours

- Restore from earlier backup version
- File share recovery from cloud
- RPO: up to 24 hrs
- RTO: few hours

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3. Primary site outage

- Failover SQL to replica server
- Restore website from cloud

- RPO: near zero
- RTO: 30–60 min

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4. Cloud outage

- Use local backup copy
- Restore systems locally
- RPO: varies
- RTO: longer (2–4 hrs)

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8) Key Trade-Offs

- Spent more on SQL replication for speed
- Saved money on file storage using slower backups
- Balanced cost vs recovery speed